

FedPE: Adaptive Model Pruning-Expanding for Federated Learning on Mobile Devices

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Abstract—Recently, federated learning (FL) as a new learning paradigm allows multi-party to collaboratively train a shared global model with privacy protection. However, vanilla FL running on heterogeneous mobile edge devices still faces three crucial challenges: communication efficiency, statistical heterogeneity, and system heterogeneity. To tackle them simultaneously, we devise FedPE, a communication-efficient and personalized federated learning framework, which allows each client to search for personalized optimal local subnets adaptive to system capacity in each round of FL. It consists of three core components: a) adaptive pruning-expanding controls model pruning or expanding according to the accuracy variations of local models, b) error compensation strategy promotes the pruned or expanded subnets to be Lottery Ticket Networks (LTNs), c) the fair aggregation rule aggregates local models with their real-time contributions as coefficients to boost the performance of the aggregated global model. The integration of the three components facilitates that only personalized optimal subnets with different footprints interact between the server and clients, which effectively reduces communication costs and enhances the robustness of FL to statistical and system heterogeneity. We also prove the convergence of FedPE and design an optimal hyperparameter searching (OHS) algorithm based on Pareto optimization to search for optimal hyperparameters for FedPE. Extensive experiments evaluated on five real-world datasets with IID or Non-IID distributions demonstrate that FedPE configured with found optimal hyperparameters achieves $1.86 \times -121 \times$ communication efficiency improvement with almost no accuracy degradation, presenting the best trade-off between model accuracy and communication cost.

Index Terms—Federated learning, communication efficiency, statistical heterogeneity, system heterogeneity, lottery ticket network.

I. INTRODUCTION

WITH the rapid development of IoT, smart mobile devices such as cell phones and wearable devices generate rich edge data, and it's popular to use these data for modeling and decision-making [1], [2], [3]. However, directly collecting

edge data to a data center has two issues: a) privacy may be leaked during collecting and storing data centrally [4], and b) transferring data consumes high communication bandwidth [5]. As a solution, federated learning (FL) [6] came into being and has been applied in various AI applications [7], [8], [9], [10], [11]. In a typical FL paradigm, clients (edge devices) train local models on local data and upload trained models to the server. The server aggregates the received local models to update the global model and broadcasts the updated global model to clients. The above steps are repeated until the global model converges or the maximum rounds are reached. In FL, only models are transmitted and data is always stored locally, which protects privacy and saves communication costs.

Whereas FL with heterogeneous mobile edge devices still faces three crucial challenges: a) *Communication efficiency*, deep neural networks often contain many redundant parameters [12], [13]. Transmitting models with whole parameters between clients and the server still consumes high network bandwidth, resulting in communication being the main bottleneck of FL on mobile edge devices with limited bandwidths [14], [15], [16], [17], [18]. b) *Statistical heterogeneity*, mobile edge devices often have unbalanced and non-independent identically distributed (Non-IID) data [19], [20]. Influenced by Non-IID, the shared single global model may not be generalized well for every client, so it's necessary to search for personalized models for each client [21], [22], [23]. c) *System heterogeneity*, mobile edge devices often have heterogeneous system resources [22], [24], [25], [26], [27], [28], [29], [30] such as computing power, bandwidth, memory, storage, etc. Low-end devices only train small models and vice versa. In vanilla FL, all clients train the same model, which constrains the model size to be afforded by the weakest client or excludes weak clients from joining in FL. Limiting model capacity or dropping data from weak clients leads to model accuracy degradation. Thus allowing devices to clip model sizes adaptive to system resources is intuitive to tackle system heterogeneity.

Existing communication-efficient FL approaches mainly include: a) increasing local training and reducing communication frequency [6], [31], b) compressing models by sparsification [14], [32], [33], [34], quantization [35], [36] and encoding [37], [38], [39], c) reducing communication latency and network congestion of communication channels [40], [41]. To deal with statistical heterogeneity, personalized FL methods [19], [42], [43] are proposed by introducing meta-learning [44], [45], multi-task learning [46], [47], transfer learning [48], [49], hybrid

Manuscript received 4 June 2023; revised 2 March 2024; accepted 5 March 2024. Date of publication 8 March 2024; date of current version 3 October 2024. This work was supported in part by the National Science Foundation of China under Grant 62272252 and Grant 62272253, in part by the Key Research and Development Program of Guangdong under Grant 2021B0101310002, and in part by the Fundamental Research Funds for the Central Universities. Recommended for acceptance by S. misra. (Corresponding authors: Gang Wang; Xiaoguang Liu.)

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Digital Object Identifier 10.1109/TMC.2024.3374706

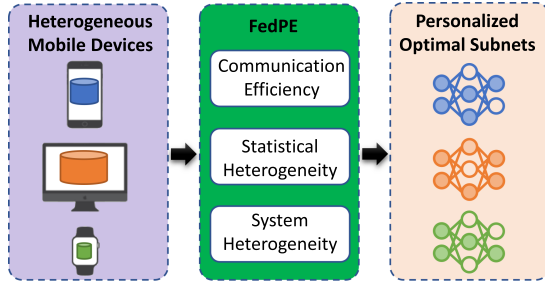


Fig. 1. High-level view of FedPE.

models [50], [51], regularization [52] etc. To tackle system heterogeneity, prior arts make each client extract heterogeneous sub-network with size tailoring to system capacity through model pruning [53], [54], knowledge distillation [55], low-rank factorization [56], etc. However, most of the above efforts are devised to solve one of the three challenges.

Allowing clients to train the subnets of the complete model by model pruning becomes a promising research line to tackle the above three challenges. However, current methods belonging to this category only can address one or two of the three challenges simultaneously. Specifically, according to the executor of pruning, these methods are divided into two groups: 1) *Pruning by the server*: In SNIP [57] and PruneFL [22], the server prunes the global model and broadcasts the same pruned global subnets to all clients, which does not address statistical and system heterogeneity since it does not consider the Non-IID data distributions and heterogeneous system capacities of clients. In FedMP [58], the server prunes the global model to different subnets afforded by heterogeneous system resources of different clients, which only improves communication efficiency and system heterogeneity. 2) *Pruning by clients*: Some methods allow clients to prune subnets only driven by local data, such as LotteryFL [59], CELL [60], Hermes [21], FedMask [61]. Although these methods tackle communication efficiency and statistical heterogeneity due to searching personalized subnets, they omit whether the pruned personalized subnets can be trained within the system capacity of heterogeneous clients or not. Some other methods allow clients to prune subnets only driven by system power, such as HeteroFL [53] and FjORD [54], they alleviate communication overheads and system heterogeneity but do not handle statistical heterogeneity. The absence of simultaneously tackling the three challenges motivates us to allow *clients* to prune *personalized optimal subnets* adaptive to Non-IID data distributions and heterogeneous system capacities.

In this work, to improve communication efficiency and robustness to statistical, system heterogeneity in FL simultaneously, we propose an adaptive pruning-expanding federated learning framework called FedPE dedicated to allowing *clients* to search for *personalized optimal subnets* with footprint matching to system capacities, as depicted in Fig. 1. There are three challenges faced by FedPE: 1) how to determine different run-time pruning or expanding ratios adaptive to heterogeneous clients? 2) how to guarantee model convergence during adaptive pruning and expanding processes? 3) how to promote the best trade-off between model accuracy and system efficiency? To tackle the

above three challenges, we devise three corresponding interlocking components. First, FedPE adaptively prunes or expands local models with pruning or expanding ratios determined by real-time accuracy variations of local models on local data. Second, to prevent the information carried by the pruned parameters permanently lost, we devise an error compensation strategy for pruning and expanding, which promotes FedPE to search for optimal subnets and ensures model convergence. Third, we develop an accuracy-driven aggregation rule to fairly aggregate local subnets for boosting the performance of the global model.

Contributions: Our contributions are summarized as:

- We propose a communication-efficient and personalized FL framework termed FedPE to tackle communication efficiency, statistical heterogeneity, and system heterogeneity simultaneously by allowing clients to search for personalized optimal subnets with footprints tailored to system capacities.
- We theoretically prove the convergence of FedPE and propose an optimal hyperparameter searching (OHS) algorithm based on Pareto optimization to enable FedPE to take the best trade-off between model accuracy and communication cost.
- Extensive experiments on five real-world datasets demonstrated the superiority of FedPE in improving communication efficiency and robustness to statistical, system heterogeneity.

Organization: The remainder of this paper is organized as follows: Section II summaries related work. Section III illustrates the preliminary studies on parameter redundancy degree, introduces the concept and merits of one-shot model pruning, and describes the background of FL. Section IV introduces our proposed FedPE framework and proves its convergence. An optimal hyperparameter searching algorithm based on Pareto optimization is devised in Section V. Section VI presents the experimental setups, results, and analysis. Section VII concludes this work.

II. RELATED WORK

A. Traditional Model Pruning

Model pruning includes weight-level, layer-level, filter-level, and image-level [62], we focus on the first one. Early pruning efforts utilize the second-order Taylor expansion of gradients, which has a high overhead for computing the Hessian matrix [63]. Later, magnitude-based pruning is proposed in [64], where parameters with magnitudes below the threshold are pruned, it tends to perform a one-shot pruning after training model to converge [65]. Iterative pruning [66] executes pruning while training model, i.e., repeating training-pruning, and the pruned subnets often perform better than one-shot pruning. Dynamic pruning [67] allows models to grow or cut during training by introducing periodic interactions of sensitive connections, i.e., repeating training-pruning-recovering. All these methods are proposed in centralized training scenarios and cannot be directly applied to FL or other parameter-server distributed deep learning architectures [68], [69].

B. Model Pruning in FL

Server Pruning with Same Subnets: In SNIP [57], the server executes one-shot pruning on the initialized global model and broadcasts it to clients, with no further pruning during subsequent federated training. PruneFL [22] involves two stages: a) initial pruning before FL: pruning the global model to a certain sparsity on a most capable and trusted client, and b) further pruning during FL: the server performs adaptive pruning on the global model at certain round intervals. Since the above methods prune the global model on the *server* to obtain the same subnet for all clients, the pruned subnets are not always optimal for all clients, which may degrade model performances. Although reducing communication costs, these methods can not tackle statistical and system heterogeneity.

Server Pruning with Different Subnets: In FedMP [58], the server adaptively prunes the global model into different subnets afforded by heterogeneous system capabilities of edge clients by solving a Multi-Armed Bandit (MAB) problem. FedDrop [70] also allows the *server* to prune the global model into several heterogeneous subnets with different pruning rates for clients by a dropout schema. They improve communication efficiency and system heterogeneity but take no account of addressing statistical heterogeneity since pruning subnets by the server does not consider the Non-IID distributions of clients' local data.

Client Pruning Driven by Data: Pruned subnets with similar accuracy to *base* (complete) models are often referred to as lottery ticket hypothesis (LTH) [71], and dense LTH subnets possibly outperform the sparse base models [59]. Inspired by LTH, LotteryFL [59] allows *clients* to search for their personalized lottery ticket networks (LTNs) of local models through pruning driven by *local data*, the server and clients always communicate personalized subnets, which effectively alleviates the Non-IID issue and improves communication efficiency. However, several critical hyperparameters of LotteryFL, such as each round's pruning rate, are manually fixed to be a constant, which is harmful to searching for the optimal subnets. To alleviate the straggler issue that may occur in LotteryFL, the server in CELL [60] broadcasts the complete global model instead of corresponding subnets to all clients on the downlink. In Hermes [21], clients exploit *structured pruning* based on *local data* to identify personalized subnets to improve communication efficiency and computation efficiency for inference. FedMask [61] lets each client learn a heterogeneous binary mask to get a personalized and structured spare submodel. Although all the above methods improve communication efficiency and statistical heterogeneity through pruning personalized subnets adaptive to local data, they do not consider system heterogeneity. The personalized subnets found by them may not be afforded by low-end devices and become stragglers.

Client Pruning Driven by System Power: In HeteroFL [53] and Fjord [54], there are different levels in the system powers of heterogeneous clients. Before federated learning, they pre-define each client's ordered and nested subnets with size tailoring to system capabilities through pruning. During federated learning, the structure of each client's subnet is fixed. In short, the subnet is determined *only by the system power* of each device,

so the pre-defined subnets are not always optimal for all clients, which may lead to model accuracy drops. Therefore, these methods only tackle system heterogeneity and communication efficiency but do not face statistical heterogeneity.

Our insights: The above existing FL model pruning methods can not address system heterogeneity, statistical heterogeneity, and communication efficiency at the same time. Unlike the above orthogonal methods, our FedPE allows *clients* to *adaptively* prune or expand local subnets based on both dynamic accuracy variation on *local data* and *system resources*, different clients always hold personalized heterogeneous optimal subnets afforded by system powers in different rounds. In stage-1 of FedPE, initially pruning local models driven by system resources allows high-end devices to hold large subnets and low-end devices to hold small subnets, which address system heterogeneity. In stage-2 federated training of FedPE, further pruning local models driven by accuracy variation on local data helps search the optimal personalized subnets adaptive to local non-iid data distributions, which alleviates statistical heterogeneity and ensures superior model performances. Since the server and clients always communicate optimal subnets, communication costs are reduced compared with communicating complete models in FedAvg. To sum up, our FedPE can address the three challenges of FL simultaneously.

C. Model Expanding in FL

Server Expanding: Model expanding has been extensively explored in continuous learning towards making the model not catastrophically forgetful for a given data series. FEWER [72] applies model expanding to FL. It initializes a sparse global model's subnet on the server, then the *server randomly* recovers several parameters to expand the model capacity of the updated global model and broadcast the *same* expanded updated global subnet to all clients in each round. The communication efficiency is improved owing to transmitting subnets, but the statistical and system heterogeneity are not solved due to the same subnets for all clients.

Our insights: Unlike recovering *random* parameters for model expanding by the *server* in FEWER, considering that the knowledge of one parameter is permanently lost once it is pruned, which is never argued in the above FL pruning methods, we devise an error compensation strategy for pruning and expanding to make each *client* to adaptively recover pruned parameters based on accumulated residuals to facilitate optimal subnet searching.

III. PRELIMINARY STUDY

In this section, we experimentally evaluate the high parameter redundancy degree of deep neural networks then introduce one-shot model pruning and FL's background.

A. Parameter Redundancy Degree

Prior work [12], [13] mentioned that most parameters of large-scale (> 100MB) deep neural networks (such as VGG, ResNet, etc.) tend to 0 and are almost useless. Neural networks deployed

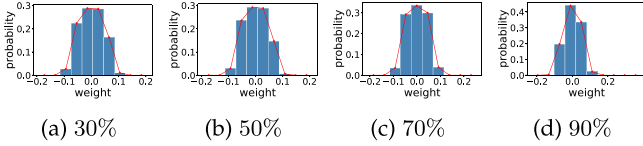


Fig. 2. Parameter distributions of LeNet when it reaches 30%, 50%, 70%, and 90% accuracy on the MNIST dataset.

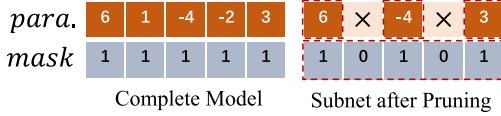


Fig. 3. Pruned Subnet denoted with masks.

on mobile edge devices are often small-scale ($< 1\text{MB}$), such as LeNet. We conduct experiments with LeNet on the MNIST dataset to observe the parameter distribution during model training. As shown in Fig. 2, LeNet also contains many parameters close to 0 at multiple training stages, indicating a high degree of parameter redundancy. Since SGD is robust to loss, dropping some parameters within the bounded-loss tolerance has almost no influence on model convergence [12]. Therefore, it's promising to clip redundant parameters to improve FL's communication efficiency.

B. One-Shot Model Pruning

Concept: Model pruning [62] is a popular approach for removing redundant parameters of models, it mainly involves types: *one-shot pruning*, *iterative pruning*, and *dynamic pruning*. One-shot pruning [57] we used in FedPE belongs to unstructured pruning techniques, it prunes the parameters with bottom- $p\%$ magnitudes at once to reduce model capacity. Specifically, the masks of the pruned parameters are set to 0 and others are still 1, only all masks and parameters masked with 1 are communicated between clients and the server, hence reducing communication costs compared with transmitting complete models. For example, as depicted in Fig. 3, the client prunes the parameters $[1, -2]$ with bottom-2 magnitudes in the complete model $[6, 1, -4, -2, 3]$, the pruned parameters are masked with 0 and the remaining parameters are masked with 1. Then the client sends masks $[1, 0, 1, 0, 1]$ and the remaining parameters $[6, -4, 3]$ of its subnet to the server, the server puts $[6, -4, 3]$ in order on the positions masked with '1' and fill 0 into the other positions masked with '0' to recover the subnet $[6, 0, -4, 0, 3]$. In short, Masks denote the structure of one subnet. After acquiring the masks and remaining parameters of one subnet, then the subnet can be recovered. Only communicating subnets like communicating complete models in FedAvg can not disclose the data privacy of clients since local data are always stored in clients locally.

Merit: Compared with the other two pruning methods detailed in Section II-A, one-shot pruning consumes the lowest computation cost. Mobile edge devices participating in federated

learning often have limited computation power, except for local model training, these devices also require paying extra computation on model pruning in FedPE. Therefore, one-shot pruning with the lowest computation complexity is the best choice for pruning local models on mobile edge devices. Besides, from the perspective of multiple continuous rounds of FedPE, each client repeats training-(one-shot) pruning-expanding steps like dynamic pruning in one round, which helps search the optimal subnet. The above two reasons motivated us to apply one-shot pruning during the whole FL process.

C. Background of Federated Learning

FedAvg [6] is a classical algorithm of vanilla FL. Suppose there are N clients in total, and the server randomly selects C fraction of clients ($C \cdot N = K$) to participate in each round of FL. The k -th client trains the local model ω_k on its local dataset D_k , and the local training objective is:

$$\min F_k(\omega_k) = \frac{1}{n_k} \sum_{i \in D_k} f_i(\omega_k), \quad (1)$$

where $n_k = |D_k|$; $f_i(\omega_k) = \ell(\mathbf{x}_i, y_i; \omega_k)$, i.e., the loss of sample $(\mathbf{x}_i, y_i)$ on model ω_k . The epoch of local training is E and the batch size is B , i.e., executing $E \frac{n_k}{B}$ local iterations. Then, clients upload trained local models to the server for aggregating, and the loss of the aggregated global model is:

$$f(\omega) = \sum_{k=0}^{K-1} \frac{n_k}{n} F_k(\omega_k), \quad (2)$$

where n is the total number of samples owned by all clients. After aggregation, the server broadcasts the updated global model ω to clients. The above processes repeat until the global model converges or the maximum round is attained.

IV. METHODOLOGY

In this section, we first formulate motivation and claim notations. Then we introduce FedPE workflow and detail its three core components: adaptive pruning-expanding, error compensation, and fair aggregation. Finally, we discuss the benefits of FedPE and prove its convergence in theory.

A. Problem Formulation and Notations

Definition: Background on the substantiated high parameter redundancy degree of deep neural networks, to simultaneously tackle the three challenges (statistical heterogeneity, system heterogeneity, and communication efficiency) in FL, we dedicate to searching for each local model ω_k 's *personalized optimal (lottery ticket) subnet* $\tilde{\omega}_k$ with footprint matching with system capabilities by allowing clients to adaptively prune or expand local models. So the problem can be formulated as

$$LTN : \tilde{\omega}_k = \text{FedPE}(\text{base model} : \omega_k). \quad (3)$$

The found personalized lottery ticket subnets (LTNs) have minimal model footprints while keeping compared accuracy with base (complete) models [59], which are beneficial to improving

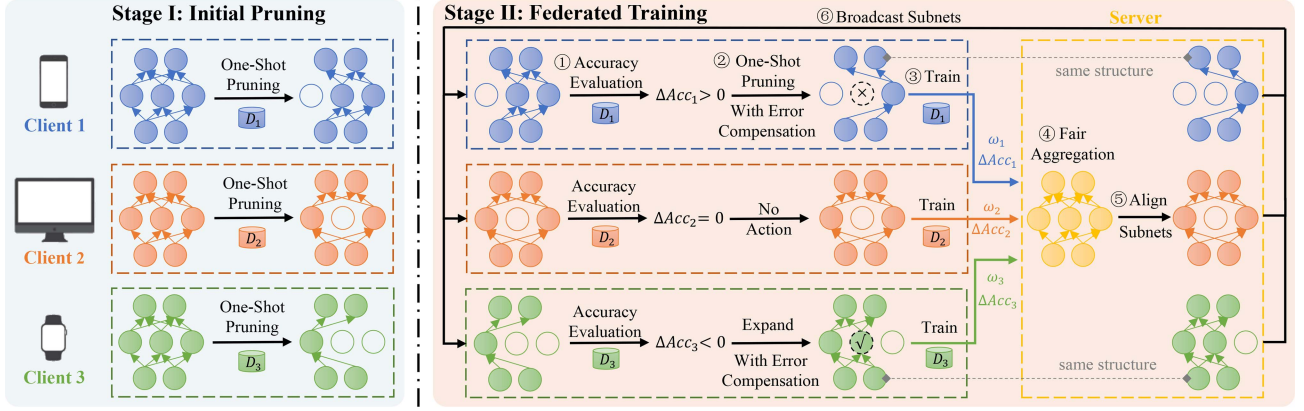


Fig. 4. Workflow of FedPE. The black dashed circle with $\times$ or $\checkmark$ indicates that the parameter is pruned or recovered.

communication efficiency and robustness to statistical, system heterogeneity.

Notations: We assume the following notations: Acc_k^t , the accuracy of the local model ω_k in the t -th round; $\tilde{\omega}_k$, the pruned or expanded subnets of the local model ω_k .

B. Overview of FedPE

To fulfill the above goal, we design FedPE which involves two stages: *initial pruning* and *federated training*. The workflow of FedPE is illustrated in Fig. 4 and the corresponding algorithm description is given in Algorithm 1.

Stage-I: Initial Pruning: For compatibility of FL with system-heterogeneous clients, before federated training, we first *initially prune* each client's complete local model with maximum parameter capacity afforded by heterogeneous system resources through one-shot pruning on local data. After initial pruning, high-end clients hold initial local subnets with large parameter capacities while low-end clients have initial local subnets with small sizes. The initially pruned subnet as the starting point of FL is further pruned or expanded in FL, and the parameter capacity of the further pruned or expanded subnet in FL always does not exceed that of the initially pruned subnet size. This ensures that all clients can train subnets adaptive to system resources in FL.

Stage-II: Federated Training: In each round: 1) The k -th client receives the personalized subnet from the server as its local model, then validates its accuracy Acc_k^{t+1} on local data and calculates the accuracy variation $\Delta Acc_k^{t+1} = Acc_k^{t+1} - Acc_k^t$ compared with the last round's accuracy Acc_k^t . 2) The k -th client decides to *adaptively prune*, *expand* or do not modify local subnets according to the accuracy variation ΔAcc_k^{t+1} . If $\Delta Acc_k^{t+1} > 0$, i.e., the accuracy is improved compared with the last round, the k -th client *prunes* the local model to reduce model capacity; if $\Delta Acc_k^{t+1} < 0$, the local model is *expanded* to grow model capacity; otherwise no extra operation is done. The pruning or expanding ratio is driven by accuracy variation on local data. The *error compensation* strategy is devised to facilitate optimal subnet searching during pruning or expanding. 3) The k -th client trains processed local subnets on local data and then uploads the masks and remaining parameters of local

subnets also with accuracy variations to the server. During local training, we freeze the pruned parameters to be 0. 4) The server aggregates received local subnets with the designed accuracy-weighted *fair aggregation* rule to update the global model. 5) The server aligns the updated global model with each received local subnet's structure, i.e., the aligned global subnets have the same coordinates of remaining parameters as received local subnets. 6) The server broadcasts the masks and remaining parameters of global structure-aligned subnets to corresponding clients as the next round's starting point. The above processes are iteratively executed until the global model converges or the maximum round is reached.

Next, we detailedly introduce the three core components in stage-II federated training of FedPE: adaptive pruning-expanding, error compensation, and fair aggregation.

C. Adaptive Pruning-Expanding

We allow *clients* to prune or expand local models driven by *accuracy variation on local data*, which has two strengths: a) the accuracy variations of different clients in one round are diverse, which enables different clients to hold *personalized* subnets adaptive to local data distribution; b) pruning or expanding models according to accuracy variations is beneficial to search the *optimal* subnets for each client in each round. The found *personalized optimal subnets* for each client on local Non-IID data in each round tackle statistical heterogeneity and communication costs effectively.

Model Pruning: As shown in Fig. 4, if accuracy variation $\Delta Acc_k^{t+1} > 0$, i.e., the accuracy of local model ω_k^{t+1} in the current round is improved than the accuracy of local model ω_k^t in the last round, the current local model may still have redundant parameters that can be pruned further, so we prune ω_k^{t+1} to reduce parameter redundancy degree further.

To guarantee no obvious accuracy degradation of local models after pruning, except for 1) $\Delta Acc_k^{t+1} > 0$, we constrain two extra conditions: 2) $Acc_k^{t+1} \geq Acc_k^{TH}$, the current model's accuracy Acc_k^{t+1} exceeds the threshold Acc_k^{TH} ($Acc_k^{TH} \in (0, 1)$), $Acc_k^{TH} > 0$ ensures that local models with not bad performances are pruned, we set $Acc_k^{TH} < 1$ because pruning local models

when Acc^{TH} approaches 1 (i.e., local models are close to convergence) leads to still communicating complete models like FedAvg during the rounds before pruning and no improved communication efficiency); 3) $pr_k^t \leq pr^{target}$ ($pr^{target} \in (0, 1)$), it determines the maximum pruned parameter capacity, we set $pr^{target} > 0$ since clients require to prune local models, we set $pr^{target} < 1$ since we cannot prune all parameters of the local model), the cumulative pruning rate pr_k^t during t rounds does not exceed the target pruning rate pr^{target} . When the above three conditions are satisfied simultaneously, *one-shot pruning* will be performed.

We define an accuracy-driven current pruning rate in the $(t+1)$ -th round as $r_k^{t+1} = \sigma(\alpha \cdot \Delta Acc_k^{t+1})$, where $\sigma(\cdot)$ is a sigmoid function that constrains the range of the current pruning rate to be $(0,1)$, α is a hyperparameter with positive value. Then we further prune the remaining parameters of the current subnet ω_k^{t+1} with $bottom-r_k^{t+1} \cdot |\omega|$ magnitudes, where $|\omega|$ is the parameter capacity of a base (complete) model. After pruning, the cumulative pruning rate pr_k^{t+1} in this round turns into

$$\text{PruningRate} : pr_k^{t+1} = pr_k^t + r_k^{t+1}, r_k^{t+1} = \sigma(\alpha \cdot \Delta Acc_k^{t+1}). \quad (4)$$

Model Expanding: If $\Delta Acc_k^{t+1} < 0$, the model accuracy in the current round decreases than that in the last round, this may be because some important parameters are pruned in previous rounds. So we require to recover these important parameters through model expanding to improve the accuracy of local models.

We utilize *residuals* (detailed in Section IV-D) to determine which parameters are recovered and define an accuracy-driven expansion rate as:

$$\text{ExpandingRate} : ex_k^{t+1} = \sigma(\beta \cdot \Delta Acc_k^{t+1}), \quad (5)$$

where β is a positive hyperparameter. Since the cumulative pruning rate pr_k^t is always bounded with pr^{target} , we do not limit the scopes of β and α . $ex_k^{t+1} \leq pr_k^t$, i.e., the parameter capacity of expanded models should not exceed that of base (complete) models. After expanding, the cumulative pruning rate is updated to be $pr_k^{t+1} = pr_k^t - ex_k^{t+1}$.

No Action: If $\Delta Acc_k^{t+1} = 0$, i.e., there are no changes in accuracy between the local models in the current round and last round, indicating that the subnet is optimal in the current round, so no action requires to be taken.

D. Error Compensation

To promote the pruned or expanded local subnets to be lottery ticket networks (LTNs) in each round, we design an *error compensation* strategy to accumulate the parameter residuals introduced by adaptive pruning-expanding.

Pruning with Error Compensation: Assuming the subnet received by the k -th client from the server in the $(t+1)$ -th round is ω_k^{t+1} and the residuals accumulated by the k -th client in the t -th round is A_k^t , similar to [33], we define the pruning process

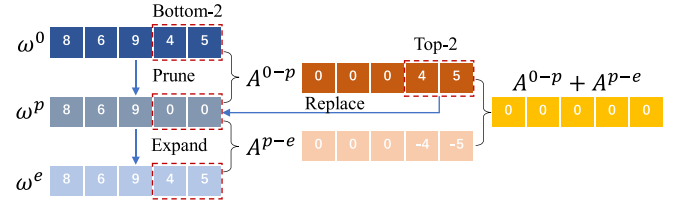


Fig. 5. Toy example of error compensation.

with error compensation as:

$$\begin{aligned} \hat{\omega}_k^{t+1} &= \omega_k^{t+1} + A_k^t, \\ \tilde{\omega}_k^{t+1} &= \text{prune}(\hat{\omega}_k^{t+1}; pr_k^{t+1}), \\ A_k^{t+1} &= A_k^t + \hat{\omega}_k^{t+1} - \tilde{\omega}_k^{t+1}, \end{aligned} \quad (6)$$

where $\tilde{\omega}_k^{t+1}$ is the pruned subnet, and the residual term $A_k^{t=0}$ is initialized to be a matrix with the same shape as local models and 0 elements. The k -th client first stacks the received model ω_k^{t+1} in the $(t+1)$ -th round with the residuals A_k^t accumulated in the t -th round as $\hat{\omega}_k^{t+1}$. Then, the k -th client prunes $\hat{\omega}_k^{t+1}$ with the cumulative pruning rate pr_k^{t+1} defined in (4). Finally, it stacks errors $\hat{\omega}_k^{t+1} - \tilde{\omega}_k^{t+1}$ introduced by pruning in this round on the historical residuals A_k^t to update A_k^{t+1} .

Expanding with Error Compensation: Instead of randomly recovering parameters like FEWER [72], we first propose to select the parameters to be recovered based on *residuals* A_k^t . Specifically, assuming that the pruned indexes of received ω_k^{t+1} is I_k^t , we sort the residual terms of indexes I_k^t by magnitudes, then select the indexes with $top-ex_k^{t+1} \cdot |\omega|$ (expanding rate $\times$ base model capacity) residual magnitudes to be recovered, and the recovered parameter values are replaced by corresponding residual terms.

For ease of understanding the significance of expanding with error compensation, we give a toy example in Fig. 5. Supposing that the initial residual vector A^0 is $\vec{0}$, we first prune model ω^0 to get ω^p , and the accumulated residuals in this step are A^{0-p} . Then, we expand ω^p to get ω^e , the accumulated residuals in this step are A^{p-e} . The stacked residuals of the above two steps are $A^{0-p} + A^{p-e}$. Since we use the values of the residual items (4 and 5 in A^{0-p}) as the recovered parameter values (4 and 5 in ω^e) during expansion, the errors caused by pruning and expanding cancel each other out.

After expansion, ω_k^{t+1} is updated to be $\tilde{\omega}_k^{t+1}$. Then we also stack the residuals introduced by expanding on A_k^t to obtain A_k^{t+1} . Meanwhile, the pruning rate pr_k^t after model expanding is updated to be pr_k^{t+1} , i.e.,

$$\begin{aligned} \tilde{\omega}_k^{t+1} &= \text{expand}(\omega_k^{t+1}; ex_k^{t+1}, A_k^t), \\ A_k^{t+1} &= A_k^t + \omega_k^{t+1} - \tilde{\omega}_k^{t+1}, \\ pr_k^{t+1} &= pr_k^t - ex_k^{t+1}. \end{aligned} \quad (7)$$

In general, benefiting from error compensation, some important parameters pruned in previous rounds are possible to be recovered during model expanding, so the contributions carried by them can be reused to boost model performance. Therefore,

adaptive model pruning-expanding configured with error compensation facilitates identifying lottery ticket networks (LTNs) of local models and guarantees model convergence. Besides, since the error compensation strategy utilizes accumulated parameter residuals during pruning to recover parameters, it can be generally coupled with other pruning methods such as dynamic pruning for parameter recovery.

E. Fair Aggregation

After training the pruned or expanded local subnets on local data, then the masks and remaining parameters of trained local subnets are uploaded to the server for aggregation. As formulated in (2), FedAvg assigns the fixed aggregation coefficient n_k/n to the k -th client, where n_k is the number of samples held by the k -th client and n is the total number of samples held by all clients. The insight of this setting is that the local model trained by the client with more data performs better. However, if the client has much but quality-poor data, the model trained on it may be worse than other models trained on less but quality-good data. So the fixed aggregation coefficients can not reflect how local models contribute to the global model in real time.

To tackle the above issue, we devise a fair aggregation rule, where the aggregation coefficients of local models are formulated by their real-time accuracy variations. Specifically, we also allow clients to upload the *byproducts* ΔAcc_k^t calculated during adaptive pruning-expanding to the server and utilize them to define the coefficients of local models as:

$$\zeta_k^t = \frac{\sigma(\Delta Acc_k^t)}{\sum_{k=0}^{K-1} \sigma(\Delta Acc_k^t)}, \sigma(\Delta Acc_k^t) = \frac{1}{1 + e^{-\Delta Acc_k^t}}. \quad (8)$$

where K is the number of clients participating in one round of FL; $\Delta Acc_k^t \in [-1, 1]$; $\sigma(\cdot)$ is a sigmoid function, mapping ΔAcc_k^t within $[0, 1]$. Then we replace n_k/n of (2) with ζ_k^t , then the new aggregation rule becomes:

$$\text{FairAggregation} : f(\omega) = \sum_{k=0}^{K-1} \zeta_k^t \cdot F_k(\omega_k). \quad (9)$$

Fairness: Models with decreased accuracy ($\Delta Acc_k^t < 0$) are allocated with small positive ζ_k^t after sigmoid mapping, and vice versa. This fair “more work, more pay” aggregation rule fully highlights the practical contributions of local models to the global model, which is beneficial to improve the global model. The global model with enhanced accuracy can indirectly increase ΔAcc_k^{t+1} , which promotes pruning in the next round. Besides, acquiring the accuracy variations of clients can not infer the local data of clients inversely, so sending accuracy variations does not violate the local data privacy of clients.

Aggregation by Ordinate. Heterogeneous subnets are denoted with different masks. After receiving the masks and remaining parameters of subnets, the server first recovers subnets and then aggregates them to generate a complete global model. When aggregating heterogeneous subnets, for each ordinate of the complete global model, the server weighted averages the parameters of subnets masked with 1 in this ordinate with the normalized accuracy variation as the coefficient to generate the parameter of the complete global model in this ordinate. Fig. 6

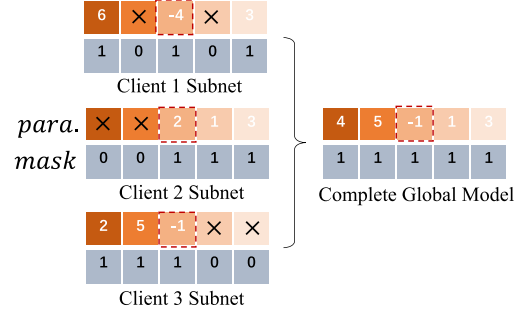


Fig. 6. Parameter aggregation by ordinate.

displays an example of parameter aggregation by ordinate, we directly average the parameters on the same ordinate for ease of understanding.

Subnet Alignment: After fair aggregation, the server uses the received *masks* of local subnets to obtain the *global subnets* with the same pruned indexes of local subnets, then broadcasts the masks and remaining parameters of the aligned global subnets to corresponding clients. That is, the distributed *global subnets* (server-to-client) have the same structures as the uploaded *local subnets* (client-to-server).

F. Discussion

The three components (adaptive pruning-expanding, error compensation, fair aggregation) of FedPE both promote to search for an optimal subnet (LTN) for each client in each round. In adaptive pruning-expanding, each client adaptively prunes or expands its local subnet driven by the dynamic accuracy variation of local subnets on local Non-IID data in consecutive two rounds, i.e., each client can adjust its subnet size according to real-time accuracy, which benefits searching for optimal subnets with satisfied accuracy in each round. In error compensation, the residuals accumulated in pruning processes are exploited to determine the indexes and values of parameters required to be recovered during expanding, it allows parameters that have been pruned to have chances to be recovered to contribute their inherent knowledge, which enables searching for optimal subnets due to subnet structure with flexible contraction and expansion. In fair aggregation, the server weight averages parameters from local subnets in the same ordinates also based on accuracy variation, which helps improve the performance of the global model and then promotes further pruning in the next round, i.e., enabling optimal subnet searching.

Owing to the found optimal personalized heterogeneous subnets by two stages, FedPE effectively alleviates the three key challenges simultaneously.

- **System heterogeneity:** The initial pruning driven by *system capacity* of heterogeneous devices limits the maximum subnet that can be trained during federated training. High-end devices have larger subnets while low-end devices have smaller subnets, which addresses system heterogeneity. In federated training, the pruned or expanded subnets in FL always has the same or smaller sizes than the initial pruned

subnets, so they are always afforded by system resources, also supporting system heterogeneity in FL.

- *Statistical heterogeneity*: The optimal subnet of each client is searched by adaptive model pruning or expanding according to dynamic accuracy variation on local data and is further trained on local data before uploading to the server, so the found optimal subnets after local training are personalized and adaptive to local Non-IID data distribution. Besides, after the server aggregates the received local personalized heterogeneous subnets to generate the complete global model, it then produces the personalized heterogeneous global subnet for each client through subnet alignment and then sends it to the corresponding client for the next round. Hence, the *personalized* heterogeneous subnets trained in FedPE enhance its robustness to statistical heterogeneity.
- *Communication efficiency*: In each round of uplink communication in FedPE, each client uploads the masks (1-bit integer values) and remaining parameters (32-bit float values) of found local optimal *subnets* and accuracy variation (one 32-bit float value) to the server. In the downlink communication, the server sends the masks and remaining parameters of the aligned global subnet to each client. In each round of communication in typical FedAvg, each client and the server communicate the whole parameters (32-bit float values) of the complete local or global model in the uplink or downlink communication. Compared with FedAvg, the communication costs consumed in FedPE are reduced significantly.

G. Computational Complexity Analysis

For the *server*, except for aggregating the received local subnets by the fair aggregation rule, it also executes the subnet alignment operation to produce heterogeneous global subnets for different clients. The introduced extra computational costs can be acceptable for the power-sufficient server.

For *clients*, training local subnets consumes lower computational overheads than training complete local models. Except for training local subnets in FL, clients execute one-shot pruning in the initial pruning stage and adaptive model pruning during federated training, and they also evaluate the accuracy of the received global subnet from the server on local data and calculate the accuracy variation between the current and last rounds. These extra computation overheads depend on model sizes. The reduction of computational overheads by training subnets and the increase of computational costs by pruning and accuracy evaluation may marginally affect or even reduce the whole computational burden because model training requires multiple iterations but pruning is executed once in each round of FL.

H. Convergence Proof

We refer to the assumptions in [73] that:

Assumption 1: The loss function of each client's local model is L -Lipschitz smooth, i.e, the loss function f satisfies:

$$\|\nabla f(X) - \nabla f(Y)\| \leq L\|X - Y\|, \quad (10)$$

Algorithm 1: FedPE.

Randomly initialize local models $[\omega_0^0, \omega_1^0, \dots, \omega_{N-1}^0] =$ global model ω^0 , then iteratively pruning local models with initial pruning rate $[pr_0^0, pr_1^0, \dots, pr_{N-1}^0]$ on local data $[D_0, D_1, \dots, D_{N-1}]$.

for each round $t=0, 1, \dots, T-1$ **do**
 $K \leftarrow \max(C \cdot N, 1)$, // C : sampling fraction, N : total number of clients
 $S_{t+1} \leftarrow$ randomly sample K clients.

Clients:

for each client $k=0, 1, \dots, K-1$ **do**

Receive ω_k^t from the server,

$Acc_k^{t+1} = \text{Evaluate}(\omega_k^t)$,

$\Delta Acc_k^{t+1} = Acc_k^{t+1} - Acc_k^t$,

if $\Delta Acc_k^{t+1} > 0$, $Acc_k^{t+1} \geq Acc^{TH}$, and $pr_k^t \leq pr_k^{target}$ **then**

$\hat{\omega}_k^{t+1} = \omega_k^t + A_k^t$,

$pr_k^{t+1} = pr_k^t + \sigma(\alpha \cdot \Delta Acc_k^{t+1})$,

/ Prune */*

$\tilde{\omega}_k^{t+1} = \text{prune}(\hat{\omega}_k^{t+1}; pr_k^{t+1})$,

$A_k^{t+1} = A_k^t + \hat{\omega}_k^{t+1} - \tilde{\omega}_k^{t+1}$,

end

else if $\Delta Acc_k^{t+1} < 0$, $Acc_k^{t+1} \geq Acc^{TH}$ **then**

$ex_k^{t+1} = \sigma(\beta \cdot \Delta Acc_k^{t+1})$,

if $ex_k^{t+1} \leq pr_k^t$ **then**

$\hat{\omega}_k^{t+1} = \omega_k^t$,

/ Expand */*

$\tilde{\omega}_k^{t+1} = \text{expand}(\hat{\omega}_k^{t+1}; ex_k^{t+1}, A_k^t)$,

$A_k^{t+1} = A_k^t + \hat{\omega}_k^{t+1} - \tilde{\omega}_k^{t+1}$,

$pr_k^{t+1} = pr_k^t - ex_k^{t+1}$,

end

end

else

/ No Action */*

$\tilde{\omega}_k^{t+1} = \omega_k^t$, $pr_k^{t+1} = pr_k^t$, $A_k^{t+1} = A_k^t$,

end

$\mathcal{B} \leftarrow$ divide D_k into batches of size B ,

for each local epoch e from 1 to E **do**

for batch $b \in \mathcal{B}$ **do**

/ Local Training */*

$\tilde{\omega}_k^{t+1} = \tilde{\omega}_k^{t+1} - \eta \nabla \ell(\tilde{\omega}_k^{t+1}; b)$,

end

end

Upload masks and remaining parameters of local subnet $\tilde{\omega}_k^{t+1}$, ΔAcc_k^{t+1} to the server.

end

Server:

Receive masks and remaining parameters of local subnet $\tilde{\omega}_k^{t+1}$, ΔAcc_k^{t+1} from K clients,

/ Fair Aggregation */*

$\omega^{t+1} = \sum_{k=0}^{K-1} \zeta_k^{t+1} \tilde{\omega}_k^{t+1}$, $\zeta_k^{t+1} = \frac{\sigma(\Delta Acc_k^{t+1})}{\sum_{k=0}^{K-1} \sigma(\Delta Acc_k^{t+1})}$,

for each client $k=0, 1, \dots, K-1$ **do**

/ Subnet Aligning */*

$\omega_k^{t+1} = \text{align_subnet}(\omega^{t+1}, \text{masks of } \tilde{\omega}_k^{t+1})$,

end

Send masks and remaining parameters of aligned global subnet ω_k^{t+1} to the k -th client.

end

Return personalized subnets $[\omega_0^{T-1}, \omega_1^{T-1}, \dots, \omega_{N-1}^{T-1}]$.

where L is a constant, X, Y are variables.

Assumption 2: The model's stochastic gradients and weights are bounded, i.e.,

$$\mathbb{E}\{\|g\|^2\} \leq \delta^2, \mathbb{E}\{\|\omega\|^2\} \leq \sigma^2. \quad (11)$$

Theorem 1: With the above assumptions, we can derive that the expectation of k -th client's gradient's $L2$ -norm within T rounds is bounded to a constant, which proves that FedPE can converge over wall-to-wall time.

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\{\|\nabla f(\tilde{\omega}_k^t)\|^2\} &\leq \frac{2(f(\omega_k^0) - f(\omega_k^*))}{\eta T} + \eta L \delta^2 \\ &+ \frac{L^2 \sigma^2 pr^{target}}{T} = \varepsilon. \end{aligned} \quad (12)$$

Proof: Based on the above assumptions, for the k -th client, we can get:

$$\begin{aligned} \mathbb{E}\{f(\omega_k^{t+1})\} &\leq f(\omega_k^t) - \frac{\eta}{2} \|\nabla f(\tilde{\omega}_k^t)\|^2 + \frac{\eta^2 L \delta^2}{2} \\ &+ \frac{\eta L^2}{2} \|\omega_k^t - \tilde{\omega}_k^t\|^2, \end{aligned} \quad (13)$$

where $\tilde{\omega}_k^t$ is the pruned/expanded model of the k -th client in the t -th round. The above conclusion has been proved in [73], [74], here we do not go into details. So we can further calculate the expectation of the $L2$ norm of gradients during T rounds, i.e.,

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\{\|\nabla f(\tilde{\omega}_k^t)\|^2\} &\leq \frac{2(f(\omega_k^0) - f(\omega_k^*))}{\eta T} + \eta L \delta^2 \\ &+ \frac{L^2}{T} \sum_{t=0}^{T-1} \mathbb{E}\{\|\omega_k^t - \tilde{\omega}_k^t\|^2\}. \end{aligned} \quad (14)$$

The k -th client's current pruning rate in the t -th round is r_k^t . As stated in [73], [74], the errors during magnitude-based model pruning satisfy:

$$\mathbb{E}\{\|\omega_k^t - \tilde{\omega}_k^t\|^2\} \leq r_k^t \mathbb{E}\{\|\omega_k^t\|^2\} \leq r_k^t \sigma^2. \quad (15)$$

Therefore, (14) can be further approximated as:

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\{\|\nabla f(\tilde{\omega}_k^t)\|^2\} &\leq \frac{2(f(\omega_k^0) - f(\omega_k^*))}{\eta T} + \eta L \delta^2 \\ &+ \frac{L^2 \sigma^2}{T} \sum_{t=0}^{T-1} r_k^t. \end{aligned} \quad (16)$$

Since we define the accumulated pruning rate during T rounds in FedPE can not exceed the target pruning rate in Section IV-C, i.e., $\sum_{t=0}^{T-1} r_k^t = pr_k^{T-1} \leq pr^{target}$, (16) can be further derived as:

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\{\|\nabla f(\tilde{\omega}_k^t)\|^2\} &\leq \frac{2(f(\omega_k^0) - f(\omega_k^*))}{\eta T} + \eta L \delta^2 \\ &+ \frac{L^2 \sigma^2 pr^{target}}{T} = \varepsilon, \end{aligned} \quad (17)$$

where all the three terms on the right side of the above inequality are constants, which indicates that the expectation of k -th

client's gradient's $L2$ -norm within T rounds is bounded, thus proving that our FedPE can guarantee the convergence of local models. $\square$

V. OPTIMAL HYPERPARAMETERS SEARCHING

The two hyperparameters α and β in FedPE affect model accuracy (measured by mAP detailed in Section VI-A) and communication efficiency (quantified by compression ratio (CR) detailed in Section VI-A). Since manually searching can only find the near-optimal pair of (α, β) and enumerating all possible hyperparameters is time-consuming, we design an optimal hyperparameter search algorithm based on *Pareto optimization*, which involves the following steps.

1) *Manually searching.* We first manually test multiple groups of (α, β) and get $\{\alpha_i, \beta_i, mAP_i, CR_i\}_{i=1}^n$.

2) *Polynomial fitting.* Then we use the results $\{\alpha_i, \beta_i, mAP_i, CR_i\}_{i=1}^n$ searched manually to construct the regression models f_a of mAP and f_c of compression ratio (CR) on (α, β) , respectively. Here, we use *sklearn.linear_model.LinearRegression()* to fit multivariate polynomials. The independent variables are α, β , so we fit binary polynomials. Taking the binary quadratic polynomial as an example, we construct the regression model as:

$$\begin{aligned} f_a(\mathbf{u}) &= b_0 + a_0 \cdot 1 + a_1 \cdot \alpha + a_2 \cdot \beta + a_3 \cdot \alpha \cdot \alpha \\ &+ a_4 \cdot \alpha \cdot \beta + a_5 \cdot \beta \cdot \beta, \\ f_c(\mathbf{u}) &= d_0 + c_0 \cdot 1 + c_1 \cdot \alpha + c_2 \cdot \beta + c_3 \cdot \alpha \cdot \alpha \\ &+ c_4 \cdot \alpha \cdot \beta + c_5 \cdot \beta \cdot \beta, \end{aligned} \quad (18)$$

where $\mathbf{u} \in \Omega$, $\mathbf{u} = [\alpha, \beta]$ are decision vectors, Ω is the decision space; $[b_0, a_0, \dots, a_5]$ and $[d_0, c_0, \dots, c_5]$ are the coefficients of fitted polynomials $f_a(\mathbf{u})$ and $f_c(\mathbf{u})$, respectively.

3) *Pareto optimization.* We aim to search for the optimal pair of (α, β) to simultaneously maximum $f_a(\mathbf{u})$ and $f_c(\mathbf{u})$, which is a multi-objective programming (MOP) problem. Finding the *Pareto front* is a common approach to solving MOP problems. All the dots $\mathbf{PF} = \{F(\mathbf{u}) | \mathbf{u} \in \mathbf{PS}\}$ on the Pareto front $\mathbf{PS} = \{\mathbf{u} \in \Omega | \mathbf{v} \in \Omega, \mathbf{v} \succ \mathbf{u}\}$ are Pareto-optimal solutions [75]. To exactly find the Pareto front, we formulate the above MOP problem as:

$$\begin{aligned} F_a(\mathbf{u}) &= -f_a(\mathbf{u}), F_c(\mathbf{u}) = -f_c(\mathbf{u}), \\ \min F(\mathbf{u}) &= \{F_a(\mathbf{u}), F_c(\mathbf{u})\}, \end{aligned} \quad (19)$$

i.e., our new goal is to find the Pareto-optimal solutions to minimize $F_a(\mathbf{u})$ and $F_c(\mathbf{u})$ simultaneously. Here, we use *Geatpy evolutionary algorithm*¹ to solve the Pareto front.

Through the above three steps, we can search for the Pareto-optimal solutions $\mathbf{PF} = \{\alpha^*, \beta^*\}$ which take the best trade-off between model accuracy and communication cost. The algorithm is detailed in Algorithm 2.

¹[Online]. Available: <http://geatpy.com/index.php/home/>

Algorithm 2: Optimal Hyperparameter Searching (OHS).

```

/* Manually Searching */
Manually test  $n$  groups of results:
 $\{\alpha_i, \beta_i, mAP_i, CR_i\}_{i=1}^n$ 
// Polynomial Fitting */
 $f_a(\mathbf{u}) = b_0 + a_0 \cdot 1 + a_1 \cdot \alpha + a_2 \cdot \beta + a_3 \cdot \alpha \cdot \alpha + a_4 \cdot$ 
 $\alpha \cdot \beta + a_5 \cdot \beta \cdot \beta$ 
 $f_c(\mathbf{u}) = d_0 + c_0 \cdot 1 + c_1 \cdot \alpha + c_2 \cdot \beta + c_3 \cdot \alpha \cdot \alpha + c_4 \cdot$ 
 $\alpha \cdot \beta + c_5 \cdot \beta \cdot \beta$ 
/* Searching for Pareto-optimal solutions by Geatpy */
 $F_a(\mathbf{u}) = -f_a(\mathbf{u}), F_c(\mathbf{u}) = -f_c(\mathbf{u})$ 
 $\mathbf{PF} \leftarrow \text{Geatpy}(F_a(\mathbf{u}), F_c(\mathbf{u}), lb = [\alpha_{lb}, \beta_{lb}], ub =$ 
 $[\alpha_{ub}, \beta_{ub}])$  //  $lb$  and  $ub$  are the lower and upper bounds
of  $\mathbf{u} = [\alpha, \beta]$ 
Return  $\mathbf{PF} = \{\alpha^*, \beta^*\}$ 

```

VI. EXPERIMENTS

We implement FedPE on the FL framework developed in [76] with Pytorch and evaluate FedPE on five datasets. All experiments are simulated with 4 NVIDIA GeForce RTX 3090 GPUs with 24 G memory.

A. Experimental Setups

Datasets and Models: We evaluate FedPE with four computer vision (CV) tasks (three image classification tasks and one objective detection task) and one natural language processing (NLP) task (one next-word prediction task).

- *LeNet on MNIST (IID)*: the MNIST² dataset [77] is IID, and we train a LeNet model on it. We assign 50 clients, and each client randomly samples 200 training instances and 50 testing instances. All clients participate in each round of FL.
- *LeNet on MNIST (synthetic Non-IID)*: we refer to the division of Non-IID dataset³ in [59] to manually divide the MNIST dataset into Non-IID pattern. We assign 50 clients, and each client has only 2 classes of instances, 200 training instances, and 50 testing instances. All clients participate in each round of FL.
- *CNN on FEMNIST (natural Non-IID)*: we train a CNN model (2Conv+1FC) on FEMNIST⁴ dataset [78]. FEMNIST is a *real-world Non-IID* dataset that contains handwritten characters from different users. We randomly select 20 user's handwritten character images and assign them to 20 clients, and the ratio of the training set and testing set is 8 : 2. All clients participate in each round of FL.
- *YOLOv3 on Street (natural Non-IID)*: Street⁵ [76] is a *real-world edge Non-IID* dataset that involves 956 images with 7 objects (table, chair, etc) collected by 20 different streets'

cameras. We merge streets with fewer images into one client and finally assign 12 clients. All clients participate in each round of FL.

- *LSTM on Stack Overflow (natural Non-IID)*: Stack Overflow⁶ is a *real-world Non-IID* large-scale dataset that involves posts from millions of users. We choose 500 users as clients and each client has posters from one user. We train an LSTM model on it to perform next-word prediction tasks. The client participation rate is set to be 0.1 in each round of FL.

Baselines: We compare FedPE with: Local Training, clients solely train local models on local data; FedAvg [6], a classical FL algorithm; pruning-based FL methods: SNIP [57], Iterative Pruning [66], PruneFL [22], LotteryFL [59], HeteroFL [53], Fjord [54], Hermes [21], FedMP [58]; expanding-based FL method: FEWER [72]; advanced personalized FL methods: FedProx [52] achieves personalization by adding the distance between the local model and the global model as a regularization term to the local model' loss, PartialFed [79] personalizes local models by mixing parameters from the global model and the local model.

Evaluation Metrics: We evaluate FedPE and baselines with the following metrics:

- *mAP*: we use mAP (mean Average Precision) to measure *model accuracy*. $mAP = \frac{\sum_{i=1}^n AP_i}{n}$, $AP_i = \frac{1}{m_i} \sum_{j=1}^{m_i} \mathbb{I}(f(x_j) = y_j)$, where Average Precision (AP) is the average prediction accuracy of each class, n and m_i are the total numbers of classes and samples of i -th class, $\mathbb{I}(\cdot)$ is an indicator function with value 1 or 0 when $\cdot$ is true or false. We evaluate the test mAP of each client's local model on its local testing dataset and calculate the mean mAP.
- *Compression Ratio (CR)*: we take the ratio between communication traffic of FedAvg and advanced FL algorithms during the overall FL process as the compression ratio. The larger compression ratio indicates higher *communication efficiency*. We calculate the average parameter capacity of uploaded local subnets to quantify communication traffic in each round.
- *Fairness*: we measure *fairness* with the *Pearson Correlation Coefficient (PCC)* between the test mAP of local models in Local Training and FedPE like [80]. The higher correlation indicates higher fairness. Local models with higher accuracy in Local Training contribute more in FL, they should be awarded more (i.e., performing higher model accuracy) in FL. We assume that in the t -th round, the test mAPs of K standalone trained local models are $\mathbf{x}^t = \{x_0^t, \dots, x_k^t, \dots, x_{K-1}^t\}$ and the test mAPs of K local models trained in FL are $\mathbf{y}^t = \{y_0^t, \dots, y_k^t, \dots, y_{K-1}^t\}$, then the PCC between $\mathbf{x}^t$ and $\mathbf{y}^t$ is calculated by:

$$R_{PCC}^t = \frac{\sum_{k=0}^{K-1} (x_k^t - \bar{\mathbf{x}}^t)(y_k^t - \bar{\mathbf{y}}^t)}{s_{\mathbf{x}^t} s_{\mathbf{y}^t}}, \quad (20)$$

²[Online]. Available: <http://yann.lecun.com/exdb/mnist/>

³[Online]. Available: <https://github.com/jeremy313/non-iid-dataset-for-personalized-federated-learning>

⁴[Online]. Available: <https://github.com/TalwalkarLab/leaf/tree/master/data/femnist>

⁵[Online]. Available: <https://dataset.fedai.org>

⁶[Online]. Available: https://www.tensorflow.org/federated/api_docs/python/tff/simulation/datasets/stackoverflow

TABLE I

IN SYSTEM-HOMOGENEOUS FL SETTINGS, THE HIGHEST MEAN MAP (%) AND CORRESPONDING COMPRESSION RATIO (CR: ×) OF FedPE AND BASELINES ON FIVE DATASETS

Methods	MNIST (IID)		MNIST (Non-IID)		FEMNIST (Non-IID)		Street (Non-IID)		Stack Overflow (Non-IID)	
	mAP	CR	mAP	CR	mAP	CR	mAP	CR	mAP	CR
Local Training	74.44	-	16.36	-	74.27	-	5.35	-	15.71	-
FedAvg [6]	98.71	1.00	93.80	1.00	86.54	1.00	60.24	1.00	23.52	1.00
SNIP [57]	98.71	1.16	97.18	1.71	85.92	1.01	55.84	1.01	21.69	1.25
Iterative Pruning [66]	98.55	1.15	96.92	1.68	86.45	1.15	56.18	1.09	22.84	1.29
PruneFL [22]	98.15	1.16	97.02	1.37	84.94	2.11	55.02	1.16	21.37	1.33
LotteryFL [59]	98.43	1.27	95.88	3.47	85.87	3.21	55.74	1.35	21.46	1.42
FEWER [72]	98.51	1.06	97.01	1.52	86.48	1.33	55.98	1.12	21.83	1.31
HeteroFL [53]	98.71	1.00	93.80	1.00	86.54	1.00	60.24	1.00	23.52	1.00
FjORD [54]	98.71	1.00	93.80	1.00	86.54	1.00	60.24	1.00	23.52	1.00
Hermes [21]	98.47	1.19	97.28	1.44	84.97	2.07	56.35	1.26	22.94	1.37
FedMP [58]	98.71	1.00	93.80	1.00	86.54	1.00	60.24	1.00	23.52	1.00
FedProx [52]	57.24	1.00	99.05	1.00	59.72	1.00	5.08	1.00	10.29	1.00
PartialFed [79]	78.20	1.00	19.77	1.00	75.83	1.00	5.77	1.00	11.33	1.00
FedPE (ours)	98.60	120.99	97.01	78.27	86.29	67.97	67.61	2.73	25.62	1.86

All results are the average of five trials.

TABLE II

IN SYSTEM-HOMOGENEOUS FL SETTINGS, THE HIGHEST COMPRESSION RATIO (CR: ×) AND CORRESPONDING MEAN MAP (%) OF FedPE AND BASELINES ON FIVE DATASETS

Methods	MNIST (IID)		MNIST (Non-IID)		FEMNIST (Non-IID)		Street (Non-IID)		Stack Overflow (Non-IID)	
	CR	mAP	CR	mAP	CR	mAP	CR	mAP	CR	mAP
Local Training	-	74.44	-	16.36	-	74.27	-	5.35	-	15.71
FedAvg [6]	1.00	98.71	1.00	93.80	1.00	86.54	1.00	60.24	1.00	23.52
SNIP [57]	36.15	93.49	36.15	88.04	38.44	52.81	1.15	50.46	1.46	20.79
Iterative Pruning [66]	13.53	93.83	19.69	90.41	10.43	83.41	1.19	52.84	1.51	21.68
PruneFL [22]	36.15	93.49	5.80	95.52	27.19	77.82	1.21	50.69	1.58	20.55
LotteryFL [59]	1.27	98.43	3.81	94.50	3.21	85.87	1.39	50.82	1.62	20.63
FEWER [72]	2.37	97.02	2.36	95.99	2.38	81.47	1.35	50.27	1.53	20.96
HeteroFL [53]	1.00	98.71	1.00	93.80	1.00	86.54	1.00	60.24	1.00	23.52
FjORD [54]	1.00	98.71	1.00	93.80	1.00	86.54	1.00	60.24	1.00	23.52
Hermes [21]	1.19	98.47	1.44	97.28	2.07	84.97	1.26	56.35	1.37	22.94
FedMP [58]	1.00	98.71	1.00	93.80	1.00	86.54	1.00	60.24	1.00	23.52
FedProx [52]	1.00	57.24	1.00	99.05	1.00	59.72	1.00	5.08	1.00	10.29
PartialFed [79]	1.00	78.20	1.00	19.77	1.00	75.83	1.00	5.77	1.00	11.33
FedPE (ours)	120.99	98.60	78.27	97.01	67.97	86.29	2.73	67.61	1.86	25.62

All results are the average of five trials.

where $\bar{x}^t$, $\bar{y}^t$ and s_{x^t} , s_{y^t} are the average values and standard deviations of x^t , y^t , $R_{PCC}^t \in [-1, 1]$.

Training Strategies: After grid search, we set the following optimal hyperparameters: local epoch $E = 10$, batch size $B = 5$, SGD optimizer with learning rate $\eta = 0.01$ for all algorithms, and $Acc^{TH} = 0.5$, $pr^{target} = 0.9$, $\alpha = 18$, $\beta = 0.1$ for FedPE.

Implementation of FedPE. First, in stage-1 initial pruning, clients prune the complete global model to heterogeneous subnets with parameter capacity afforded by heterogeneous system resources (storage and memory) through one-shot pruning on local data. Then, in stage-2 federated training, the initially pruned subnets as starting points of FL are further pruned or expanded adaptively according to the pruning or expanding rate determined by the accuracy variation between the current and last rounds on local data. At the end of FL, each client obtains its personalized heterogeneous optimal subnet used for model performance evaluation.

B. Comparisons With Baselines

We test FedPE and baselines with different hyperparameters in an FL scenario with system-homogeneous devices since only

a few baselines HeteroFL, FjORD, FedMP support system heterogeneity. In a homogeneous FL system, we assume that all clients can train complete models, so the three methods degrade to FedAvg. We report the highest model accuracy with the corresponding compression ratio and the highest compression ratio with the corresponding model accuracy of personalized local models.

1) *Mean mAP:* Tables I and II show that FedPE achieves the highest compression ratios while presenting compared or improved accuracy with FedAvg (no communication optimization) on all datasets, outperforming baselines. This substantiated the effectiveness of FedPE in balancing statistical heterogeneity (Non-IID) and communication efficiency.

We display the variations of model accuracy and communication traffic (the mean parameter capacity of transmitted subnets) with rounds on the FEMNIST dataset in Fig. 7(a), (b) (matching with Table I) and Fig. 7(c), (d) (corresponding to Table II). We observe that FedPE converges to similar accuracy to FedAvg while consuming the lowest communication cost (showing the smallest area between the curve and axes). The clear mean subnet size variations of FedPE during federated training in Fig. 8 show that the mean subnet size swells or shrinks during training and stabilizes eventually.

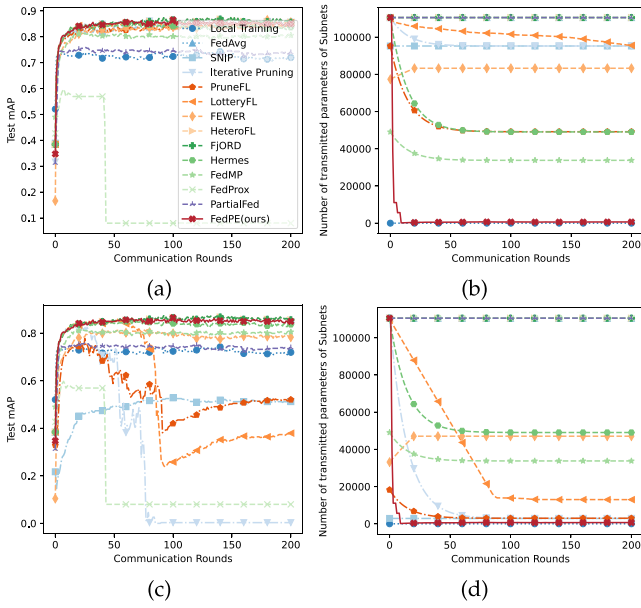


Fig. 7. Model accuracy and communication traffic (the mean parameter number of transmitted subnets) vary with rounds when FedPE and baselines achieve (a)–(b): the highest mAP, (c)–(d): the highest compression ratio on FEMNIST dataset, respectively. The area between the curve and axes in (b) and (c) is the overall communication traffic during 200 rounds.

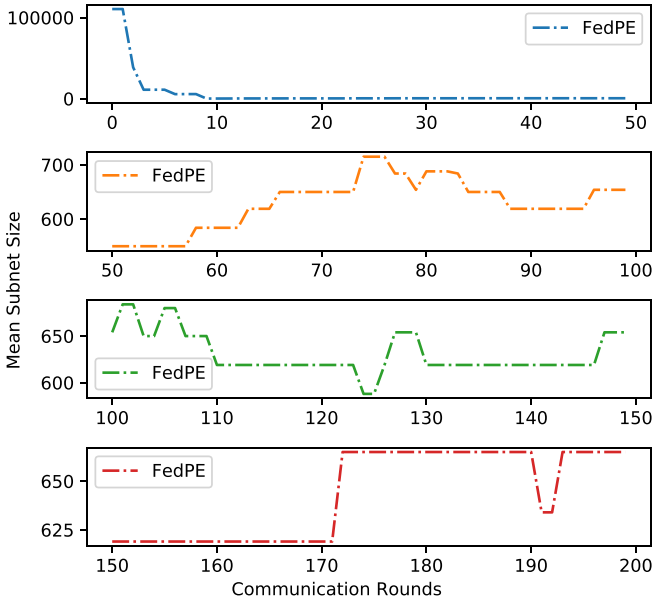


Fig. 8. Mean subnet size (parameter capacity) of FedPE varies as communication rounds on the FEMNIST dataset.

In FedProx, each client adds a regularization measured by the distance between the global model and the local model on the local training loss, which induces the local model towards the global model. Although it alleviates the non-IID issue, the communication cost is the same as FedAvg. Fig. 7(a) and (c) present that FedProx converges to about 60% accuracy and then diverges in the subsequent rounds, because only adding a regularization produces limited accuracy improvements, and

local models may be trapped into over-fitting in the subsequent training rounds.

The expanding-based FEWER has higher accuracy but lower communication efficiency than pruning-based algorithms. Although pruning-based baselines reduce communication costs, the model accuracy presents obvious degradation since pruning models with fixed hyperparameters or other special pruning rules can not identify optimal subnets. Especially, Fig. 7(c) depicts that the three pruning methods Iterative Pruning, PruneFL, and LotteryFL present obvious model accuracy degradation after they converge to the highest accuracy during further pruning in the subsequent rounds. Iterative Pruning repeats training and pruning without definite a pruning stop condition. PruneFL first prunes the global model to a certain sparsity and then the server further prunes the sparse global model in several intervals of rounds, also without a pruning stop condition. LotteryFL allows each client to prune its personalized heterogeneous subnet with a fixed pruning rate in one round, and local models are continuously pruned until reaching the target pruning rate. When the three algorithms converge to the highest model accuracy, no or a high pruning stop condition forces them to further prune models, and the continuously reduced model size leads to an obvious model accuracy drop.

FedPE's each round's current pruning or expanding rate is defined by accuracy variation. When models are close to converging, the small accuracy variation determines that model size will not change obviously, so enabling stable model accuracy. The above results reflect the necessity of considering the real-time model accuracy variations during pruning or expanding for searching optimal subnets in FedPE.

2) *Individual mAP*: We also report the test mAPs, the test mAP margins compared with Local Training, remaining parameter numbers and compression ratios of 20 individual personalized subnets in the last round of FedPE and baselines when reaching the highest compression ratio on the FEMNIST dataset (i.e., consistent with Table II).

Except for FedAvg with no communication optimization (mean mAP: 86.54%), FedPE presents the darkest pixels in Fig. 9(a) and the lightest pixels in Fig. 9(b), i.e., overall mAPs of personalized local models in FedPE outperform baselines. This demonstrates that FedPE has found optimal subnets (i.e., LTNs). Since LotteryFL also allows clients to adaptively prune local models with a fixed current pruning rate, it reaches personalized models' mAPs (mean mAP: 85.87%) second only to FedPE (mean mAP: 86.29%).

In Fig. 10, the final personalized local models in our FedPE have the lowest parameter capacity and the highest compression ratio, which reflects that FedPE is the most communication-efficient. It can be seen that the personalized local models of FedPE (Fig. 10(b)) and LotteryFL, Hermes (Fig. 10(a)) have subnets with diverse parameter numbers since clients determine to prune local models adaptively. In SNIP, iterative pruning, PruneFL, FEWER, local models share the same model structure since the server prunes or expands the global model and then sends it to all clients. Local Training, FedAvg, FedProx, PartialFed, and

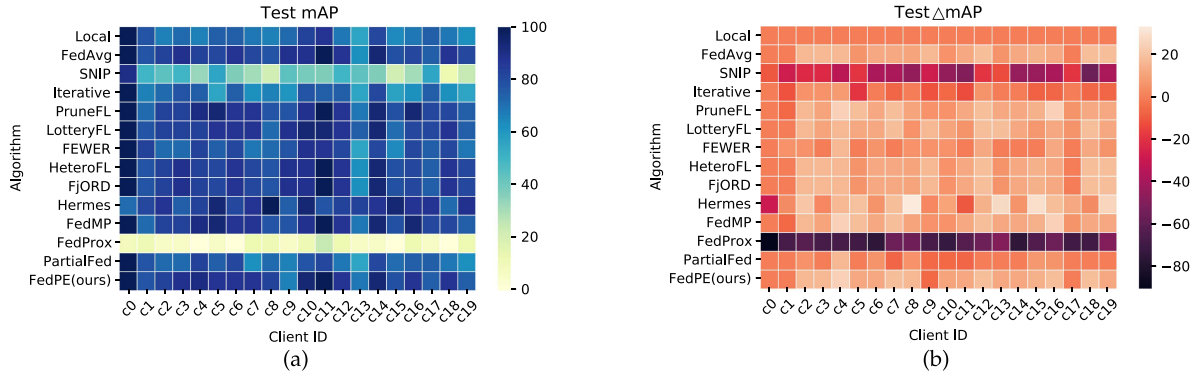


Fig. 9. On the FEMNIST dataset, (a): the mAP (%) of personalized local subnets in the last round of baselines and FedPE, (b): the margin between the mAP of personalized local models in the last round of each algorithm and Local Training.

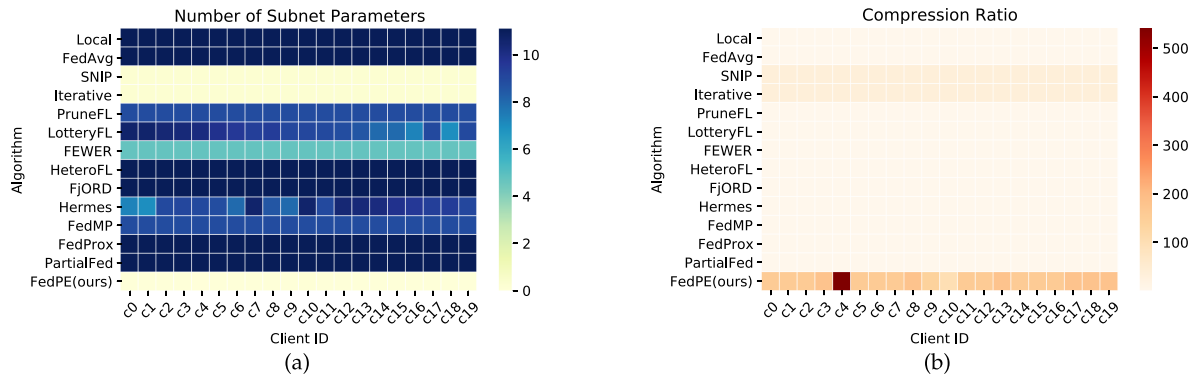


Fig. 10. On the FEMNIST dataset, (a): the remaining parameter numbers (unit: ten thousand) and (b): the compression ratio (unit: $\times$) of personalized local subnets in the last round of baselines and FedPE.

HeteroFL, FjORD, FedMP degrading to FedAvg take no operations on parameter capacity, so clients always hold base (complete) models.

C. Case Study

To explore the robustness of FedPE to different client participation rates, total numbers of clients, and system heterogeneity, we conduct case studies of FedPE and related advanced baselines: FedAvg, LotteryFL, FEWER and state-of-the-art algorithms to solve system heterogeneity: FjORD, HeteroFL, FedMP on the FEMNIST dataset. We also test the sensitivity of FedPE to hyperparameters.

1) *Robustness to Client Participation Rate*: We randomly select written characters from 50 writers in the FEMNIST dataset and assign them to corresponding 50 clients. To explore how various client participation rates affect FedPE and baselines, we vary the client participation rate $fraction = \{0.2, 0.4, 0.6, 0.8, 1.0\}$. Fig. 11(a) presents that FedPE always maintains similar mAP to FedAvg and outperforms LotteryFL and FEWER under different fractions. We can also see that the mAP of all methods improves as the fraction increases, which is intuitive since more participating clients in one round contribute more information. Fig. 11(b) displays the ratio of final personalized subnets' mean capacity and base model capacity

varies with different fractions. It's obvious that FedPE shows the lowest subnet capacity ratio, indicating that FedPE is the most communication-efficient. We can also observe that the final subnet footprint increases as the fraction rises in FedPE and LotteryFL, which may be because more biased Non-IID data held by more participating clients cause the test mAP of the global model is lower than the pruning threshold ($Acc^{TH} = 0.5$ in both FedPE and LotteryFL) in the first several FL rounds, then pruning fewer parameters. The results in Fig. 11(c)-(d) present lower pruning rates of FedPE and LotteryFL in the first several rounds under higher fractions. Since FedAvg takes no communication optimization and the local model capacity in FEWER is determined by the server, the client participation rate does not affect their final local model capacity.

2) *Robustness to Total Number of Clients*: To explore how different total numbers of clients affect FedPE and baselines, we randomly select written characters from $\{10, 20, 30, 40, 50\}$ writers in FEMNIST dataset and assign them to corresponding $\{10, 20, 30, 40, 50\}$ clients, while setting the client participation rate $fraction = 1.0$ (i.e., all clients join in aggregation). Fig. 12 depicts that FedPE always has similar or higher model accuracy compared with baselines while achieving personalized subnets with the lowest capacity under various total numbers of clients, verifying the robustness of FedPE to total numbers of clients. Besides, as the total number of clients rises, model accuracy

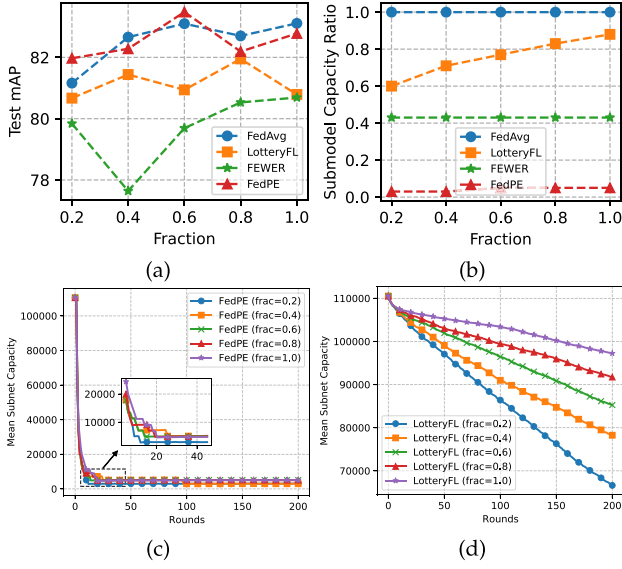


Fig. 11. On the FEMNIST dataset, (a)–(b): the test mAP and submodel capacity ratios of FedPE and baselines vary with the client participation rate (fraction). (c)–(d): the mean subnet capacity of FedPE and LotteryFL vary with rounds.

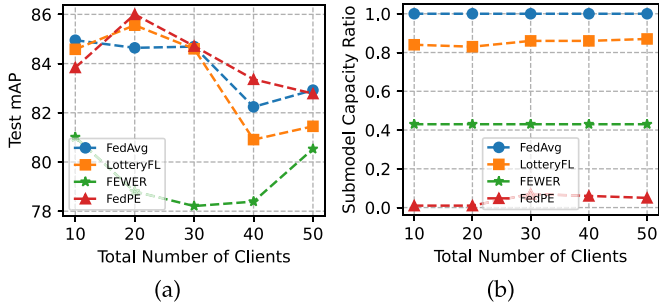


Fig. 12. On the FEMNIST dataset, the test mAP and submodel capacity ratios of FedPE and baselines vary as the total number of clients.

decreases and submodel capacity increases, which may also be due to more biased Non-IID data supplied by more clients.

3) *Robustness to System Heterogeneity*: We randomly select written characters from 50 writers in the FEMNIST dataset and assign them to 50 clients, while setting the client participation rate $fraction = 1.0$. The system heterogeneity can be quantified with clients' initial pruning rates. The powerful clients can train and communicate with larger models and vice versa, so we set lower initial pruning rates for powerful clients and higher initial pruning rates for weak clients. We compare FedPE and baselines (LotteryFL with performance second only to FedPE in the above system-homogeneous FL scenario, HeteroFL, FjORD and FedMP supporting system heterogeneity) in two system-heterogeneous settings: a) *No system heterogeneity*: We set all clients with an initial pruning rate of 0, i.e., the initial local models of all clients are base models. b) *Random system heterogeneity*: We randomly assign initial pruning rates $\{0, 0.1, \dots, 0.9\}$ to 50 clients with 4 random seeds.

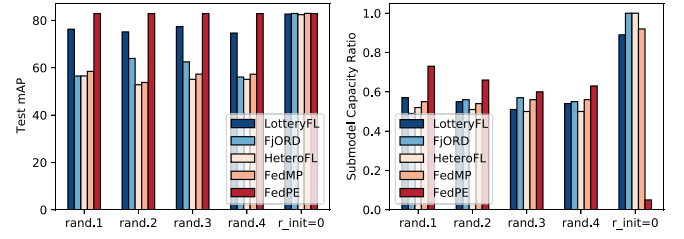


Fig. 13. On the FEMNIST dataset, the test mAP and submodel capacity ratios of FedPE and baselines under different system capability. We randomly assign initial pruning rates $\{0, 0.1, \dots, 0.9\}$ to all clients with four random seeds (2020, 2021, 2022, 2023, denoted as rand. 1–4) to simulate system heterogeneity, and we also set all clients' initial pruning rates to be 0 (denoted as $r_{init}=0$) to simulate a homogeneous FL system without system capacity differences.

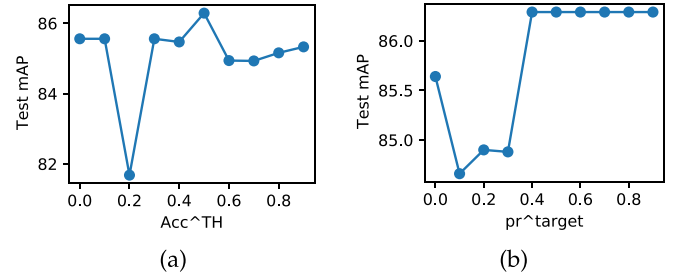


Fig. 14. On the FEMNIST dataset, the mean test mAP of FedPE varies as Acc^{TH} and pr^{target} , respectively.

Fig. 13 shows that under random system heterogeneity (case-b), FedPE achieves similar final subnet capacity but always has the highest model accuracy compared with baselines, LotteryFL follows, which not only demonstrates the robustness of FedPE to various system heterogeneity but also reflects that adaptively pruning parameters by clients (FedPE, LotteryFL) can identify better subnets than pre-defined subnets in FjORD, HeteroFL and the subnets pruned by the server in FedMP. Besides, in a homogeneous FL system without system heterogeneity (case-a), FedPE performs the highest model accuracy and the lowest model footprint, again verifying the superiority of FedPE in tackling statistical heterogeneity and communication efficiency.

4) *Sensitivity to Hyperparameters*: There are four hyperparameters in FedPE: Acc^{TH} and pr^{target} that control clients whether to prune or not, α and β which determine the pruning or expanding ratio. During comparisons with baselines, we have tested FedPE with different pairs of (α, β) and found the best-performed pairs of $(\hat{\alpha} = 18, \hat{\beta} = 0.1)$, the details are described in Section VI-D1. So we fix $(\alpha = 18, \beta = 0.1)$ and follow the setups with 20 clients on the FEMNIST dataset in comparison experiments, then we evaluate how the mean test mAP varies as: a) different $Acc^{TH} = \{0, 0.1, \dots, 0.9\}$ (bounded by $Acc^{TH} \in (0, 1)$) when pr^{target} is fixed to be 0.9 to find the best Acc^{TH} and b) different $pr^{target} = \{0, 0.1, \dots, 0.9\}$ (bounded by $pr^{target} \in (0, 1)$) with the found best Acc^{TH} to find the optimal pr^{target} .

Fig. 14(a) shows that $Acc^{TH} = 0.5$ achieves the highest model accuracy, so we fix $Acc^{TH} = 0.5$ in Fig. 14(b) and

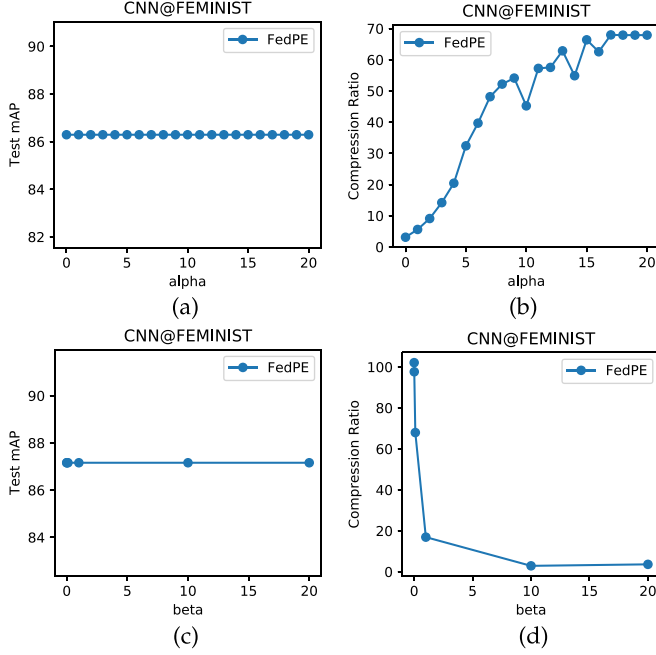


Fig. 15. On FEMNIST, (a)-(b): the results of FedPE with different α and fixed $\beta = 0.1$; (c)-(d): the results of FedPE with different β and fixed $\alpha = 18$.

observe that $pr^{target} \geq 0.4$ takes the highest model accuracy. Since the larger target pruning ratio allows clients to prune more parameters, which benefits from reducing communication costs, we consider $Acc^{TH} = 0.5$, $pr^{target} = 0.9$ as optimal hyperparameters.

D. Ablation Study

To explore the separate effects of the three core components (adaptive pruning-expanding, error compensation, fair aggregation) in FedPE, we conduct ablation studies on the natural Non-IID FEMNIST dataset.

1) *Hyperparameters of Adaptive Pruning-Expanding:* α and β are two key hyperparameters in adaptive pruning-expanding. We first fix $\beta = 0.1$ and evaluate FedPE with multiple α to find the best $\hat{\alpha}$; then we fix $\alpha = \hat{\alpha}$ and test the performances of FedPE with multiple β .

The results in Fig. 15(a) and (c) present that different pairs of (α, β) have slight effects on model accuracy, since considering error compensation in model pruning-expanding ensures model performances.

In Fig. 15(b), the compression ratio of FedPE tends to be stable when $\alpha > 15$, which indicates each client has found its *optimal* and stable subnet. As defined in (4), both the value of hyperparameter α and accuracy variation ΔAcc affect each round's pruning ratio. As α increases, the compression ratio improves, which may be because α has a greater impact on the pruning ratio and the increased α promotes a higher pruning ratio. For those points ($\alpha = \{11, 13, 15\}$, $\beta = 0.1$) with decreased compression ratios, a larger α with a smaller ΔAcc may also lead to a lower pruning ratio, so the final compression ratio decreases compared with other points. Besides, the curve inflection point

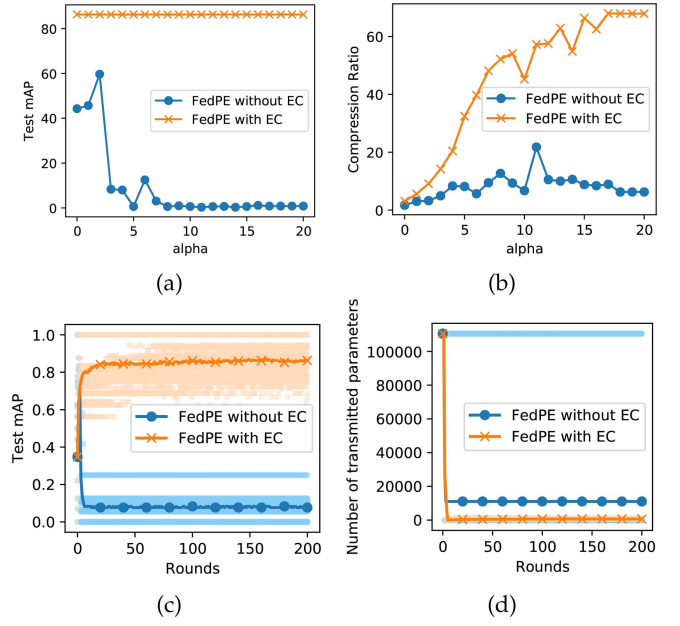


Fig. 16. On FEMNIST, (a) and (b) show the mean mAP and compression ratio of personalized models in FedPE with or without error compensation (abbreviated as EC); (c) and (d) show the *mean* (lines) and *individual* (dots) results in FedPE with or without EC when setting $(\alpha = 18, \beta = 0.1)$.

at $(\hat{\alpha} = 18, \hat{\beta} = 0.1)$ can be regarded as the near-optimal solution, in which FedPE takes the best trade-off between model accuracy (86.29%) and communication efficiency (compression ratio: $67.97\times$). So we fix $\alpha = 18$ in Fig. 15(d) and see that the compression ratio decreases as β rises, which is because the larger β formulates the higher expanding ratio, thus model capacity is recovered more.

In short, the compression ratio is sensitive with both the hyperparameters α, β and the accuracy variation ΔAcc before *optimal* subnets are found, so we should set a best-performed pair of the hyperparameters α and β to facilitate the best trade-off of FedPE between model accuracy and communication cost.

2) *With or Without Error Compensation:* We further evaluate the effect of error compensation (EC) in FedPE. We test FedPE with or without error compensation and report the results in Fig. 16. From Section VI-D1, we conclude that the larger β harms compression ratios for fixed α , so we fix $\beta = 0.1$ in this experiment.

From Fig. 16(a), we see that utilizing error compensation achieves higher model accuracy than not using it. Besides, FedPE with error compensation has a smooth variation in accuracy as α increases, while FedPE without error compensation has an obvious accuracy drop as α rises, which indicates that error compensation can effectively ensure model convergence. Fig. 16(b) displays that FedPE with error compensation has a higher compression ratio than without it. The above two observations verify that FedPE facilitates to search for optimal subnets (LTNs) with the lowest parameter capacity and satisfying model accuracy.

We also depict the test mAP and remaining parameter capacity variations of personalized models as rounds when FedPE set

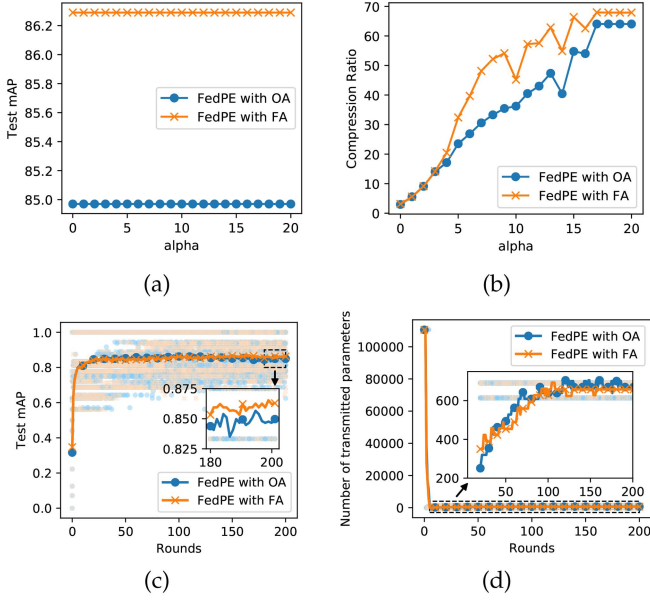


Fig. 17. On FEMNIST, (a) and (b) show the mean test mAP and compression ratios of personalized models in FedPE with fair aggregation (FA) or original aggregation (OA) rules; (c) and (d) show the mean (lines) and individual (dots) performances of FedPE configured with $(\alpha = 18, \beta = 0.1)$ and FA or OA.

with $(\alpha = 18, \beta = 0.1)$ (substantiated to be near-optimal in Section VI-D1) in Fig. 16. From Fig. 16(c), we find that individual personalized models in FedPE without error compensation have almost 0 mAP, showing the necessity of designing EC to guarantee model convergence. Fig. 16(d) shows that all personalized local models' subnets found by FedPE with EC have lower parameter capacity. The two results again verify the significance of EC to search for LTNs.

3) *Fair or Original Aggregation*: We evaluate FedPE with fair aggregation (FA) ((9)) or original aggregation (OA) ((2)) rules and report the results in Fig. 17. Similarly, we fix $\beta = 0.1$ in this experiment.

In Fig. 17, FedPE with fair aggregation (FA) performs higher accuracy and compression ratios. Since both FedPE (FA) and FedPE (OA) utilize error compensation, their accuracy variations are smooth. The fair aggregation rule also presents higher compression ratios, since FA improves model accuracy (i.e., increasing ΔAcc_k^t) and indirectly promotes the next round's pruning.

Fairness Evaluation: Fig. 18(b) presents that FedPE (FA) has a slightly higher correlation (fairness) than FedPE (OA) with standalone training, which explains why FA (assigning aggregation coefficients with the “more pay for more work” principle) achieves marginally higher accuracy than OA, as shown in Fig. 18(a). Besides, we can see that the correlations between FedPE and Local Training decrease as rounds rise, and the possible reason is that standalone training models gradually converge to *local optima* while models trained through FL approach *global optima* due to collaborative learning.

The above measurements demonstrate that fair aggregation is helpful and necessary to improve model performances and communication efficiency simultaneously.

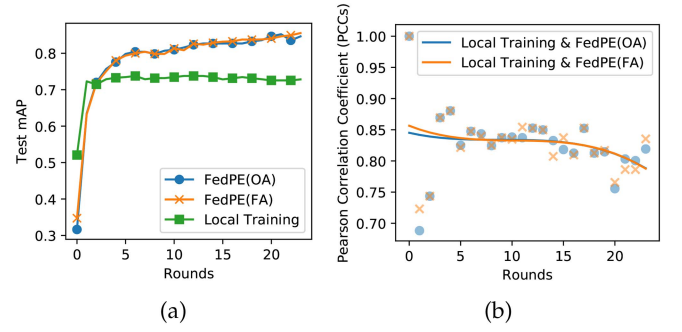


Fig. 18. On FEMNIST, (a): the average mAP of local models in Local Training and FedPE with OA/FA; (b): the correlation between Local Training and FedPE with OA/FA.

TABLE III
RESULTS OF ABLATION STUDY IN FedPE ($\alpha = 18, \beta = 0.1$) ON THE FEMNIST DATASET

Case	PE	EC	FA	mAP	CR
1	✓			7.72	9.24
2	✓	✓		84.97	64.05
3	✓		✓	0.81	8.93
4	✓	✓	✓	86.29	67.97

PE: adaptive pruning/expanding, EC: error compensation, FA: fair aggregation.

TABLE IV
COEFFICIENTS OF FITTED BINARY QUADRATIC POLYNOMIALS $f_a(\mathbf{u})$, $f_c(\mathbf{u})$ WITH DATA IN FIG. 15(A)-(B)

b_0	a_0	a_1	a_2	a_3	a_4	a_5
86.29	0.000e+00	-7.12e-30	-1.40e-45	3.23e-31	-7.12e-31	-7.36e-62
d_0	c_0	c_1	c_2	c_3	c_4	c_5
-9.03	0.000e+00	7.89e+00	8.89e-16	-2.11e-01	7.89e-01	8.45e-32

4) *Summary of Ablation Study*: To observe the numerical results of FedPE in the ablation study more clearly, we set FedPE with near-optimal $(\alpha = 18, \beta = 0.1)$ and record results in Table III. Comparing case 1 with case 2, we can see that EC effectively improves model accuracy and compression ratio simultaneously, since EC promotes PE to search for optimal subnets (LTNs). From case 2 and case 4, we find that FA further improves model accuracy and compression ratio at the same time, since FA indirectly facilitates pruning due to boosting model performances. Combining case 1/3 and case 2/4, we observe that FA must be used in conjunction with EC to be effective because FA makes sense only when models converge. Overall, all three components are crucial to enable FedPE's best trade-off between model accuracy and communication cost.

E. Optimal Hyperparameters Searched by OHS

We use the manually tested results in Fig. 15(a)-(b) to fit binary quadratic polynomials $f_a(\mathbf{u})$ and $f_c(\mathbf{u})$, and the coefficients of fitted $f_a(\mathbf{u})$ and $f_c(\mathbf{u})$ are recorded in Table IV. We present the visualized effects of polynomial fitting in Fig. 19(a)-(b). Then we put $F_a(\mathbf{u}) = -f_a(\mathbf{u})$ and $F_c(\mathbf{u}) = -f_c(\mathbf{u})$ into the Geatpy tool to search for the Pareto front, and we set $\alpha \in [1, 30], \beta \in [0, 0.1]$. The iterative results of Geatpy are displayed in Fig. 19(c).

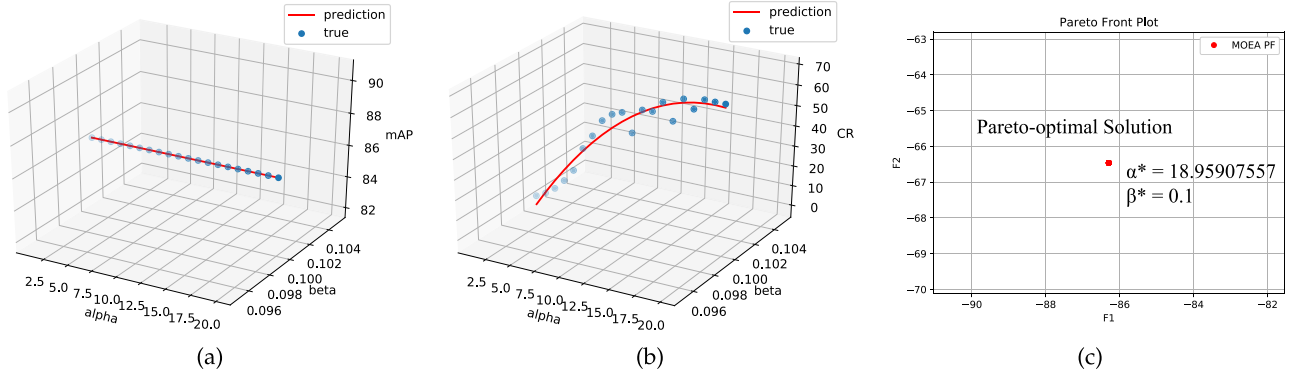


Fig. 19. (a)-(b): the visualized effects of polynomial fitting, (c): the iterative results of Geatpy on the data in Fig. 15(a)-(b).

TABLE V
THE RESULTS OF FedPE WITH PARETO-OPTIMAL $[\alpha^*, \beta^*]$ SEARCHED BY
OHS AND NEAR-OPTIMAL $[\hat{\alpha}, \hat{\beta}]$ TESTED MANUALLY

Hyperparameters	mAP	CR
OHS: $[\alpha^* = 18.95907557, \beta^* = 0.1]$	86.29	67.99
Manual: $[\hat{\alpha} = 18, \hat{\beta} = 0.1]$	86.29	67.97

As shown in Fig. 19(c), Geatpy only searched for one Pareto-optimal solution: $[\alpha^* = 18.95907557, \beta^* = 0.1]$. Since we only use data with $\beta = 0.1$ setting in Fig. 15 and Fig. 19(a) shows mAP is not affected by α , then the problem is simplified to the relation between compression ratio (CR) and α . Fig. 19(b) shows CR (α) is a monotone function, so there exists only one optimal solution.

Reasonableness Verification: We evaluate the effects of FedPE with the Pareto-optimal solution $[\alpha^* = 18.95907557, \beta^* = 0.1]$ searched by OHS algorithm and the near-optimal solution $[\hat{\alpha} = 18, \hat{\beta} = 0.1]$ searched by manually testing, and the results are reported in Table V. We can see that the mAP and compression ratio (CR) achieved by the two settings are similar, which verifies the reasonableness of the Pareto-optimal solution searched by the OHS algorithm.

VII. CONCLUSION

This paper proposed a communication-efficient and personalized FL framework called FedPE to tackle communication efficiency, statistical heterogeneity, and system heterogeneity simultaneously. It consists of three essential components: a) adaptive pruning-expanding dynamically adjusts the pruning or expanding rate to search for optimal personalized local sub-networks (LTNs), b) error compensation auxiliary searches LTNs and ensures their convergence, c) fair aggregation highlights real-time contributions of local models to improve the performance of the global model. Theoretical analysis proved that FedPE can converge. Extensive experiments demonstrated that FedPE achieved state-of-the-art communication efficiency with almost no degradation in model accuracy, surpassing existing pruning/expanding-based FL algorithms and advanced personalized FL methods. Besides, case studies on various system capacities verified the superiority of FedPE in tackling

system heterogeneity. In practice, users can utilize the developed OHS algorithm to search for optimal hyperparameters to make FedPE take the best trade-off between model accuracy and communication efficiency.

ACKNOWLEDGMENT

The authors also thank the suggestions given by reviewers and editors for improving this work.

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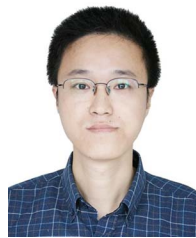
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